

Lib le Lib

Secure Dating and Support Platform for People Living with HIV

Product & Technical Build Documentation

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Prepared as an internal planning and engineering reference document.

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1. Executive Summary

Lib le Lib is an exclusive, verified mobile application designed for people living with HIV (PLHIV) to pursue marriage, serious relationships, and friendship in an environment free of disclosure-related fear and stigma. Access is restricted to members who complete a confidential medical verification process; once verified, members browse, match, and message other verified members without needing to disclose or discuss their status as a precondition of connection.

This document consolidates the full product vision, the complete feature set — including the verification, privacy, and support capabilities discussed and refined throughout planning — the technical architecture, and a step-by-step process for building, securing, launching, and operating the application responsibly. It is intended as a working reference for product, engineering, legal, and operations as the project moves from concept to build.

2. Problem Statement & Vision

2.1 Problem Statement

People living with HIV frequently face stigma, fear of rejection, and privacy risk when seeking relationships through general-purpose dating platforms, which assume disclosure of HIV status as a precondition for honest connection. This discourages many PLHIV from pursuing relationships at all, or exposes them to discrimination and emotional harm when they do disclose.

2.2 Vision

To be the most trusted, secure, and dignified relationship platform for people living with HIV, where verified status removes the need for individual disclosure and members can focus on genuine compatibility.

2.3 Objectives

- Provide an exclusive, verified space for PLHIV to pursue marriage, relationships, and friendship without disclosure-related fear.
- Protect identity and health data through privacy-first technical design, not policy alone.
- Operate strictly as a dating and matchmaking product with curated support content — not an open social or community platform.
- Build member trust through rigorous verification, clear safety tooling, and accountable moderation.

3. Target Users & Use Cases

3.1 Primary Users

- HIV-positive men and women seeking marriage, serious relationships, or friendship.

3.2 Relationship Goals Supported

- Marriage
- Serious relationship
- Friendship

3.3 User Roles

- Member — creates a profile, browses, matches, messages, reports, and blocks.
- Verification Officer — reviews medical document submissions; approves, rejects, or requests resubmission.
- Moderator — reviews reports, investigates flagged threads, and enforces community standards.
- Admin — manages system configuration, oversight, and audit review.

3.4 Core User Journeys

- New member: sign up → verification submission → pending (limited, blurred browsing) → approved → full access.
- Daily user: browse → like/pass → match → chat (photos blurred until reveal is authorized) → optional safe meetup.
- Verification Officer: review queue → approve / reject with reason / request resubmission → set re-verification date.
- Moderator: reports queue → investigate flagged thread → warn / suspend / ban → logged action.

4. Complete Feature Set

4.1 Onboarding & Verification

- Phone or email sign-up with OTP confirmation.
- Medical document submission via camera capture or file upload.
- Optional telehealth-linked verification through partnered clinics, as an alternative to self-submitted documents.
- Liveness selfie check to confirm the person verifying is the person using the account.
- Verification status lifecycle: Submitted → In Review → Approved / Rejected (with reason) → Expiry → Re-verification.
- Automated re-verification reminders ahead of document expiry (e.g., every 12–24 months).
- Resubmission flow for rejected or expired verifications, with rate limits to deter repeated fraudulent attempts.

4.2 Profile & Identity Privacy

- Nickname-only identity — no legal name field anywhere in the product.
- Age, gender, city/region, relationship goal(s), bio, and interest tags.
- Photos blurred by default; the original image is withheld from other members until a mutual match (and, optionally, explicit per-person reveal approval).
- Revocable reveal — a member can withdraw photo visibility after sharing.
- Discreet mode — hides online/activity status and read receipts, and limits push-notification content.

4.3 Discovery & Matching

- Filter by age range, gender, region, and relationship goal.
- Like / Pass / Mutual match with match notifications.
- Relationship-intent compatibility quiz (timeline expectations, disclosure preferences to family, views on children) to improve match quality for marriage-track members.
- Grid or swipe-style browsing.

4.4 Messaging & Connection

- One-to-one, end-to-end encrypted chat.
- Photo sharing inside chat using the same blur/reveal and revocation rules as profile photos.
- Optional short pre-meetup video verification call to confirm a match's identity before meeting in person.

4.5 Safety & Trust

- Block — mutual disappearance; a blocked member cannot tell they were blocked.

- Report with structured reason categories (harassment, fake profile, outing threats, solicitation, scam, underage suspicion).
- Escalation tiers and a fully audit-logged moderation process.
- In-app meetup safety checklist (share trip details externally, meet in public first).

4.6 Support & Wellbeing

- Admin-curated, professionally vetted resource library (treatment information, U=U education, hotlines) — static content, not user-generated.
- Anonymous, one-to-one Q&A access to a verified health professional.
- Opt-in, anonymized success stories used for in-app encouragement and external destigmatization messaging.

4.7 Accessibility & Reach

- Low-bandwidth / data-light mode for users with inconsistent connectivity.
- Multi-language support, starting with Amharic and English.

4.8 Admin & Verification Dashboard

- Verification queue with service-level timers.
- Moderation queue with severity tagging.
- Full audit log of every verification and moderation decision.
- Aggregate, anonymized usage metrics only — no dashboards expose individual health data beyond what review requires.

5. System Architecture

5.1 Data Model

Verification data is kept in a logically and physically separate store from the rest of the application data.

Entity	Key Fields	Notes
User	id, phone/email, credentials, status, created_at	No legal name stored.
Profile	nickname, age, gender, region_code, goals[], bio, tags[]	Public-facing data only.
Photo	profile_id, storage_ref, visibility_rule, blurred_default	Visibility resolved per request.
VerificationRecord	user_id, status, submitted_at, decision_at, reviewer_id, expiry_date	Isolated, encrypted store.
Match	user_a_id, user_b_id, matched_at, status	Created on mutual like.
Message	match_id, sender_id, encrypted_payload, sent_at, revoked	End-to-end encrypted payload.
Report	reporter_id, reported_id, category, evidence_ref, status	Drives moderation queue.
AuditLog	actor_id, action, target_id, timestamp	Every verification/moderation decision.

5.2 Privacy & Security Architecture

Concern	Approach
Chat confidentiality	End-to-end encryption (e.g., libsignal, or self-hosted Matrix).
Location	Region/city-level only; exact coordinates never stored or transmitted.
Photos	Server-side blur; original withheld until consented reveal.
Verification documents	Isolated storage, short retention window, role-gated signed-URL access.
Notifications	Generic content by default; no app name or message preview on lock screen.
App identity	Optional discreet alternate icon/name.

Analytics	First-party, aggregated, anonymized only — no third-party ad/analytics SDKs.
Access control	Role-based access control, least privilege, logged and alertable.
Data deletion	Full deletion of profile, photos, messages, and verification metadata within a fixed SLA.
Incident response	Documented breach-response plan prepared before launch.

5.3 Roles & Permissions

Action	Member	Verif. Officer	Moderator	Admin
View own profile/chats	Yes	No	No	No
View verification documents	No	Yes (queue only)	No	No
Approve/reject verification	No	Yes	No	No
View reported chat content	No	No	Yes (flagged only)	No
Suspend/ban accounts	No	No	Yes	Yes
View audit logs	No	No	Partial	Yes
System configuration	No	No	No	Yes

5.4 Technology Stack

- Mobile app: React Native or Flutter (single codebase across platforms).
- Backend: Node.js (NestJS) with PostgreSQL as the primary data store.
- Verification document storage: isolated encrypted object storage (e.g., S3 with KMS-managed keys), separate from the primary database.
- Chat: self-hosted Matrix or a custom WebSocket service using libsignal for genuine end-to-end encryption.
- Push notifications: FCM / APNs with generic, non-revealing payloads.
- Hosting region: selected per target market with attention to applicable data-residency requirements.

6. Step-by-Step Build Process

The following sequence takes the project from foundation to a live, operating product. Steps 5 and 6 run in parallel once architecture is in place; all other steps are sequential or overlapping as noted.

Step 1 — Discovery & Legal Foundation

Estimated duration: 2–3 weeks

- Finalize target launch markets and a jurisdiction strategy (some jurisdictions may need to be excluded for member safety).
- Engage a registered IP/patent agent and a lawyer experienced with health-data regulation.
- Decide the intellectual property protection mix: trademark for the brand, copyright for code/content, trade secret for internal verification logic, and patent only for genuinely novel technical mechanisms.
- Draft an initial data handling and retention policy.

Step 2 — Requirements & Specification

Estimated duration: 2 weeks

- Finalize the full feature list and prioritize into MVP, Phase 2, and Phase 3.
- Write a detailed functional specification per module.
- Define the data model and API contracts.
- Write the verification acceptance-criteria document: accepted document types, rejection criteria, and expiry period.

Step 3 — Design

Estimated duration: 3–4 weeks

- Wireframe onboarding, verification, profile, discovery, chat, and the photo blur/reveal flow.
- Establish a visual design system with a discreet, dignified tone.
- Complete an accessibility and low-bandwidth design pass.
- Run a usability review with representatives from the target community before development begins.

Step 4 — Architecture & Infrastructure Setup

Estimated duration: 2 weeks

- Stand up source control and CI/CD pipelines.
- Provision cloud infrastructure: primary database, isolated verification storage with KMS-managed encryption, and chat service infrastructure.
- Configure role-based access control and audit logging at the infrastructure level from day one.

Step 5 — Core Backend Development

Estimated duration: 6–8 weeks

- Authentication service (OTP, session management).
- Profile and discovery service.
- Matching engine and filters.
- Chat service with end-to-end encryption.
- Verification service: submission handling, review queue, status lifecycle, expiry automation.
- Admin and reporting APIs.

Step 6 — Core Mobile App Development**Estimated duration:** 8–10 weeks (parallel with Step 5)

- Onboarding and verification submission flow.
- Profile creation and editing.
- Discovery/browse interface.
- Chat interface, including blur/reveal controls and revocable photo sharing.
- Settings: discreet mode, notification preferences, account deletion.

Step 7 — Verification System Build-Out**Estimated duration:** 3–4 weeks

- Reviewer dashboard for Verification Officers.
- Liveness-check integration.
- Telehealth partner integration, if pursued.
- Re-verification reminder automation.

Step 8 — Safety & Moderation Tooling**Estimated duration:** 2–3 weeks

- Block and report flows.
- Moderator dashboard with severity tagging and escalation tiers.
- Audit-log surfaces for both verification and moderation actions.

Step 9 — Security Hardening & Compliance Review**Estimated duration:** 3–4 weeks

- Internal review of encryption, storage isolation, and access controls.
- Third-party penetration test.
- Legal compliance review per target jurisdiction.
- App store policy pre-check with Apple and Google.

Step 10 — Quality Assurance & Community Testing**Estimated duration:** 3–4 weeks

- Unit and integration testing.

- Security regression testing.
- Closed beta with real members of the target community, with direct feedback channels.
- Iterate based on beta feedback before public launch.

Step 11 — Pre-Launch Preparation

Estimated duration: 2 weeks

- Finalize the Privacy Policy and Terms of Service with legal counsel.
- Recruit and train Verification Officers and Moderators.
- Prepare a support/helpdesk channel.
- Finalize app store listings using careful, non-stigmatizing language.

Step 12 — Launch

Estimated duration: Phased, ongoing

- Begin with a phased rollout in a focused city or region to validate operations before wider release.
- Monitor verification turnaround time, safety reports, and system performance closely during early rollout.

Step 13 — Post-Launch Operations & Iteration

Estimated duration: Ongoing

- Continuously monitor safety metrics and verification service-level agreements.
- Maintain regular feedback loops with the user community.
- Roll out Phase 2 and Phase 3 features per the roadmap.
- Conduct periodic security re-audits and manage the re-verification cycle.

7. Roadmap & Timeline

Phase	Scope	Approx. Duration
MVP	Auth, manual verification, profile, discovery, matching, E2E chat with blur/reveal, block/report, admin dashboard, static resource library.	~4–5 months
Phase 2	Telehealth-linked verification, anonymous health-professional Q&A, compatibility quiz, discreet-mode polish, low-bandwidth mode, subscription billing.	+2–3 months
Phase 3	Pre-meetup video verification, multi-language expansion, refined re-verification automation, success-story tooling.	+2–3 months
Ongoing	Post-launch operations, security re-audits, and community-informed iteration.	Continuous

8. Team Structure

- Product lead.
- 1–2 mobile engineers (fewer if using a single cross-platform codebase well).
- 1–2 backend engineers.
- 1 security/privacy engineer (part-time acceptable at MVP stage, but non-negotiable given the data category).
- 1 designer with sensitivity to the target community's needs.
- Contract health-data lawyer and registered IP agent, engaged per market.
- 1–2 trained Verification Officers and 1 Moderator (can be the same trusted person initially, split as the platform grows).
- Optional: telehealth partnership coordinator if pursuing clinic-linked verification.

9. Risk Register

Risk	Mitigation
Data breach exposing HIV status	Isolated verification storage, end-to-end chat encryption, independent security audit and penetration test before launch.
Fraudulent verification	Liveness check, trained reviewers, resubmission rate limits.
Catfishing or impersonation after verification	Periodic re-verification, photo liveness checks, accessible reporting tools.
Legal liability in restrictive jurisdictions	Per-market legal review and deliberate geographic exclusions where needed.
Stigma reproduced inside the app	Context-specific moderation guidelines and clear community standards.
Regulatory or app-store rejection	Early conversations with Apple/Google review teams about data handling before full build.
Low early trust or adoption	Phased rollout, partnerships with established PLHIV support organizations, community-informed design from Step 3 onward.

10. Monetization Strategy

- Subscription tiers only — e.g., a free tier with limited daily likes, and a paid tier with unlimited matching, see-who-liked-you, and advanced filters.
- No third-party ad networks: the risk of ad platforms inferring health status from app usage outweighs any ad revenue in this category.
- Avoid pay-to-be-seen mechanics that could pressure members into reducing their privacy settings for visibility.

11. Legal & Compliance Checklist

- Confirm every jurisdiction of operation and the applicable HIV-confidentiality and data-protection laws in each.
- Engage a registered agent for trademark filing of the "Lib le Lib" name and logo with the Ethiopian Intellectual Property Authority (EIPA), and in any other relevant jurisdiction.
- Note that Ethiopia's Inventions (Patent) Proclamation No. 123/1995 excludes business methods from patentability — plan IP strategy around trademark, copyright, and trade secret protection as the primary tools, reserving patent filings for genuinely novel technical mechanisms only.
- Draft a Privacy Policy and Terms of Service specific to health-data handling, reviewed by legal counsel rather than adapted from generic templates.
- Establish a data-breach notification plan that meets the strictest timeline among all applicable jurisdictions.
- Pre-clear health- and dating-app category requirements with Apple App Store and Google Play before submission.

Appendix A — Intellectual Property Strategy Summary

Most of the durable value in this project sits in execution rather than in a single patentable idea. The recommended protection mix is:

- Trademark — the "Lib le Lib" name and logo, filed with EIPA (renewable seven-year terms).
- Copyright — source code, UI design, and written content, protected automatically upon creation under Proclamation No. 410/2004; no filing required.
- Trade secret — internal verification workflow details and matching logic that are not clearly novel; keeping these confidential is often more valuable than disclosing them in a patent application likely to be refused.
- Patent — reserved for specific, genuinely novel technical mechanisms only (for example, a particular progressive image-disclosure method or a particular privacy-preserving notification method) — not for the underlying business concept of restricting access to verified members, which is excluded as a business method under Ethiopian patent law.

A registered Ethiopian patent agent should review any claims before filing to confirm which, if any, meet the novelty and inventive-step requirements.